

Data Management Portal

API Manual

Guide to the portal REST API, with endpoints and examples

Version updated with the Jobs module ([/api/jobs](#))

Internal document — valid for the current version of the portal

Table of Contents

Table of Contents	2
1. Introduction to the API	3
1.1 Base URL	3
1.2 Response format	3
1.3 Authentication	3
1.4 Permissions	3
1.5 Status codes	3
2. Authentication and session	3
2.1 Login	3
2.2 MFA (e-mail second factor)	4
2.3 Change password	4
2.4 Logout	4
2.5 Current user	4
3. Data modules	4
3.1 Generic CRUD pattern	4
3.2 Available modules	5
3.3 Example — create a job	5
3.4 Special fields	5
4. Option lists (/tipos)	5
5. Dashboard	6
6. Reports	6
7. Administration	6
7.1 Users	6
7.2 E-mail and project	6
8. Audit and security	6

1. Introduction to the API

The portal exposes its own HTTP/JSON API (manual routing in PHP, no framework) that serves both the portal front-end and external scripts (integration routines, automation, data extraction). This manual documents the available routes, including the **Jobs** module.

1.1 Base URL

All routes live under the /api path, from the address where the portal is installed. If the portal is in a subfolder (e.g. http://server/data), the prefix becomes /data/api. The examples below show only the part from /api onward.

1.2 Response format

Responses are JSON. Errors follow the format { "detail": "message" } with the appropriate HTTP code. Dates are strings in YYYY-MM-DD format.

1.3 Authentication

The API uses a JWT token (HS256). Log in at POST /api/auth/login and send the token on every other call in the Authorization: Bearer <token> header. The token expires (8 hours by default) and can be revoked by logout, password change or an admin reset.

1.4 Permissions

Besides the token, each route may require a minimum role (Read, Write, Administrator or Master) or access to a specific module:

- **Read:** GET only, on allowed modules.
- **Write:** GET/POST/PUT/DELETE on allowed modules.
- **Administrator:** full access, including Users, Register and Settings.
- **Master:** everything the Administrator can do, plus the Audit trail.

1.5 Status codes

Code	Meaning
200	Success.
400 / 422	Invalid request (missing or malformed data).
401	Not authenticated (token missing, invalid, expired or revoked).
403	Authenticated but without permission for the action/module.
404	Route or record not found.
429	Temporary block due to too many login attempts.
500	Internal server error.

2. Authentication and session

2.1 Login

POST /api/auth/login — body with username and password (JSON or form). Without MFA it returns the token; with MFA it returns an mfa_token for the next step.

```
POST /api/auth/login
{ "username": "john", "password": "*****" }
-> { "access_token": "", "token_type": "bearer",
    "must_change_password": false }
-> or { "mfa_required": true, "mfa_token": "" }
```

If must_change_password is true, the user may only call /auth/change-password, /auth/me and /auth/logout until the password is changed — other routes return 403.

2.2 MFA (e-mail second factor)

- **POST /api/auth/mfa/verify** (/auth/mfa/verificar) — body { mfa_token, codigo }; returns the access_token.
- **POST /api/auth/mfa/resend** (/auth/mfa/reenviar) — body { mfa_token }; resends the code (max 5 resends, 60s cooldown).

2.3 Change password

POST /api/auth/change-password — body { senha_atual, nova_senha }. In the temporary-password flow it accepts { force: true, nova_senha } without the current password. Revokes the current token on completion.

2.4 Logout

POST /api/auth/logout — revokes the current token immediately.

2.5 Current user

GET /api/auth/me — logged-in user's data. **PUT /api/auth/me** — updates name, e-mail and preferences (theme, style, MFA).

3. Data modules

3.1 Generic CRUD pattern

All data modules share the same set of endpoints, changing only the base path. Replace {mod} with the module path (table below):

Endpoint	Description / permission
GET /api/{mod}	Lists records. Accepts ?q=search and ?inicio=YYYY-MM-DD&fim=YYYY-MM-DD (range). Requires read on the module.
POST /api/{mod}	Creates a record (JSON body with the module fields). Requires write.
GET /api/{mod}/{id}	Gets a record by id. Requires read.
PUT /api/{mod}/{id}	Updates a record. Requires write.
DELETE /api/{mod}/{id}	Deletes a record. Requires write.

Endpoint	Description / permission
DELETE /api/{mod}/lote	Bulk delete. Body { "ids": [1,2,3] }. Requires write.

3.2 Available modules

Module	Base path ({mod})
Access	/api/acessos
Changes	/api/mudancas
Backup	/api/backup
Restore	/api/restore
Data Dictionary	/api/dicionario
Integrations	/api/integracoes
Jobs & Routines	/api/jobs

3.3 Example — create a job

POST /api/jobs with JSON body. The comando field stores the steps as JSON text (a list of { nome, comando }):

```
POST /api/jobs
{
  "nome": "Daily DW load",
  "tipo": "ETL / SSIS",
  "servidor": "SRV-ETL01",
  "frequencia": "Daily",
  "horario": "once a day",
  "status": "Active",
  "responsavel": "BI Team",
  "comando": "[{\"nome\": \"Step 1 - Extract\", \"comando\": \"EXEC dbo.usp_extract\"}]"
}
```

3.4 Special fields

- **jobs.comando:** JSON text with the list of steps, each { nome, comando } (both free text). May be sent empty.
- **jobs.agendamentos:** JSON text with the list of time windows, each { inicio, fim } in HH:MM. May be sent empty.
- **mudancas.objetos:** sent as a JSON array of { nome, tipo }; stored in a dedicated sub-table and returned in the objetos field.

4. Option lists (/tipos)

Manages the option lists used in the forms (Type/Status/Criticality etc.), including the Jobs ones (job_tipo, job_frequencia, job_criticidade, job_status). Administrator routes:

- **GET /api/tipos** — lists all options, grouped by category.
- **POST /api/tipos** — creates an option { categoria, nome }.
- **PUT /api/tipos/{id}** — renames an option.
- **DELETE /api/tipos/{id}** — removes an option.

5. Dashboard

- **GET /api/dashboard/stats** — aggregated indicators (KPIs), including jobs_total, jobs_ativos and jobs_desativados.
- **GET /api/dashboard/export** — all modules (including jobs) in a single JSON.
- **GET /api/dashboard/info** — public environment info (project name, database engine).

6. Reports

POST /api/relatorios/{mod}/email — e-mails the module's report (PDF). Body { destino, inicio?, fim? }. Works for any module, including /jobs. Requires read on the module and a configured SMTP server.

7. Administration

7.1 Users

- **GET /api/usuarios** — lists accounts. **POST /api/usuarios** — creates an account.
- **PUT /api/usuarios/{id}** — updates (role, password, active, modules). Resetting the password revokes the user's tokens and forces a change.
- **PUT /api/usuarios/{id}/permissoes** — adjusts the allowed modules.
- **PATCH /api/usuarios/{id}/ativo** — enables/disables the account.
- **DELETE /api/usuarios/{id}** — removes the account.

7.2 E-mail and project

- **GET/PUT /api/config/email** and **POST /api/config/email/testar** — SMTP configuration and test.
- **GET/PUT /api/config/projeto** — project name, logo and session timeout.

8. Audit and security

- **GET /api/auditoria** — audit trail (Master only). Filters: usuario, tabela, acao, inicio, fim, pagina.
- **GET /api/seguranca/stats** — statistics of refused login attempts (Admin/Master).

Creations, updates and deletions in the Jobs module are automatically recorded in the audit trail, like the other modules.